CASE STUDY

Warehouse Management (WM) III

This case study explains an integrated warehouse management process, which is triggered released/set off by a sales order for material, to be delivered to the customer from a warehouse-managed storage location.

Product

S/4HANA 2023
Global Bike

Fiori 3.0

Level

Beginner

Focus

Warehouse Management

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Version

4.3

Last Update

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MOTIVATION

Warehousing has significant value for logistics.

Current trends such as higher cost pressure, shorter cycles of innovation, higher customer expectations and globalization of markets make great demands on companies, particularly on warehouse logistics. This is especially difficult in industries with high distinction like the consumer goods industry. Furthermore, customers have increasingly higher demands on reliability, promptness and flexibility of deliveries.

Warehouse management systems support the global flow of goods between the producer and the purchaser and facilitate near fail-proof logistic operations in increasingly complex supply chains.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to work successfully through this case study, it is not necessary to finish the WM exercises. Anyway, it is recommended.

NOTES

This case study uses the Global Bike (GBI) data set, which has been exclusively created for SAP UA global curricula.

SAP University Alliances

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Process Overview

Learning Objective Understand and perform a warehousing sales process cycle.

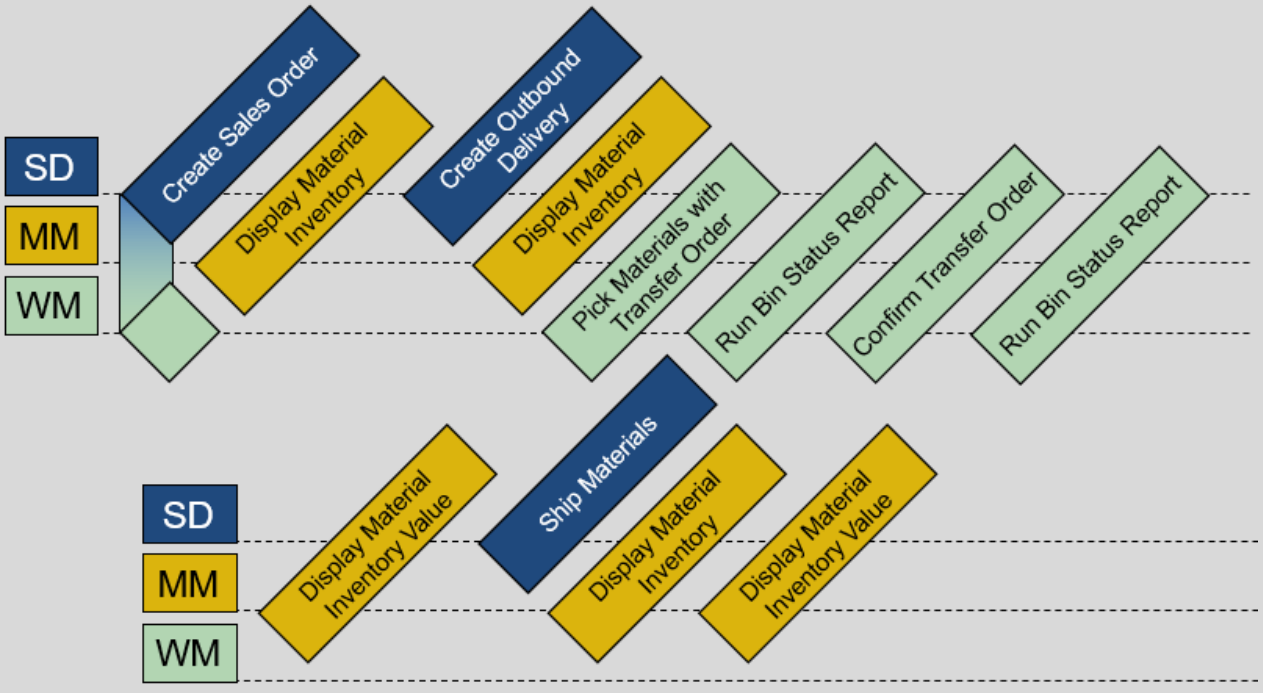
Time 85 min

Scenario Due to increasing sales output in your San Diego distribution center, the management has decided to install a Warehouse Management System there. After running some tests, you have some material in your warehouse and you can fulfill a new sales order.

Employees involved

Karim Messalem (Sales Person 1 US West)
Carolin Bruzik (Warehouse Supervisor)
Sunil Gupta (Warehouse Employee)
Zarah Morello (Good Issue Clerk)

First, you have to create a sales order. In order to fulfill this new sales order, you need to create an outbound delivery. You may remember this part from the SD case study. However, this time the process is slightly different, because you deliver from a warehouse-managed storage location. After picking the materials, goods are shipped to the customer. Because this case study focuses on Warehouse Management, detailed instructions of how to create the invoice and how to receive the payment are not included. However, you may use respective parts of the Sales and Distribution (SD) case study to finalize the sales process and see the financial impact.



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Table of Contents

Process Overview 2
Step 1: Create Sales Order 4
Step 2: Display Material Inventory 6
Step 3: Create Outbound Delivery 8
Step 4: Change Outbound Delivery 9
Step 5: Display Material Inventory 11
Step 6: Pick Materials with Transfer Order 13
Step 7: Run Bin Status Report 15
Step 8: Confirm Transfer Order 18
Step 9: Run Bin Status Report 20
Step 10: Display Material Inventory Value 22
Step 11: Ship Materials 23
Step 12: Display Material Inventory 25
Step 13: Display Material Inventory Value 26
WM III Challenge 27

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Step 1: Create Sales Order

Task Create a sales order. **Time** 10 min

Short Description Use the Fiori Launchpad to create a sales order.

Name (Position) Karim Messalem (Sales Person 1 US West)

To create a sales order, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Sales Person* role the app *Manage Sales Orders-Version 2.*

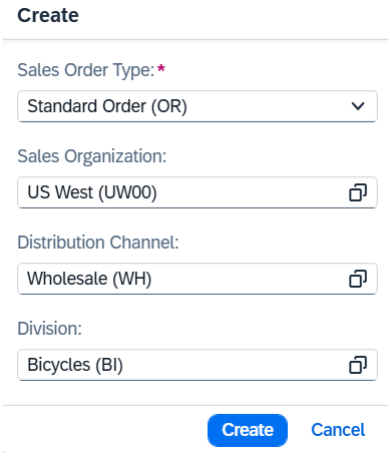
Start



In the *Manage sales orders – Version 2* view, first click on the button Create.

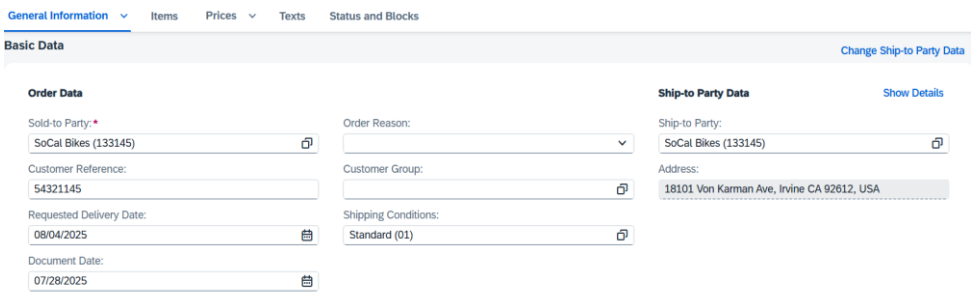
In the *Create* enter **OR** (Standard Order) as *Sales Order Type*, **UW00** as *Sales Organization*, **WH** (Wholesale) as *Distribution Channel*, **BI** (Bicycles) as *Division*. Confirm your entries by clicking Create.

OR
UW00
WH
BI



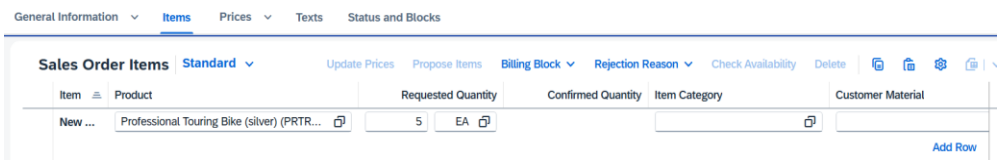
In the *Sales Order* view, enter **133###** (SoCal Bikes Irvine; replace ### with your number) as Sold-To Party and **54321###** as the *Customer Reference*. Enter **one week from today** as the *Requested delivery date*.

133###
54321###
one week from today

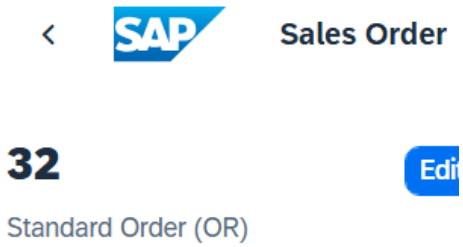


Then enter **PRTR2###** as the *product* and **5** as the *Requested quantity* in the Items area.

PRTR2###
5



Then click on Create to save your order. The system will now assign an individual order number.



Standard Order Number

Click on SAP to return to the SAP Fiori Launchpad.

Step 2: Display Material Inventory

Task View the inventory of your material. **Time** 5 min

Short Description Use the Fiori Launchpad to display the inventory of your material.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

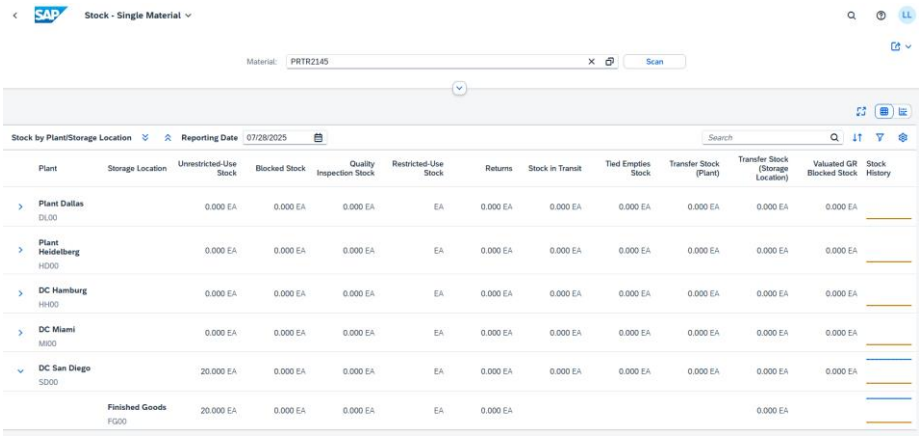
To display the material inventory, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Stock - Single Material.*

Start

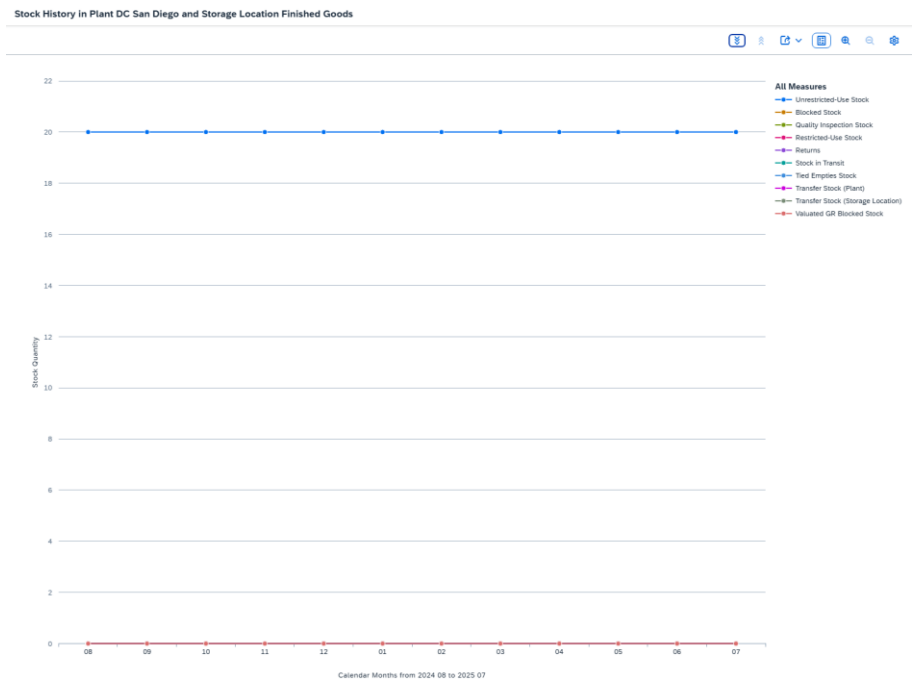


In the *Stock - Single Material* screen, enter **PRTR2###** as Material (remember to replace ### with your number). Klick Enter.

PRTR2###



Note that you still have the same amount of unrestricted-use stock in San Diego.



Click on SAP to return to the SAP Fiori Launchpad.

Step 3: Create Outbound Delivery

Task Create an outbound delivery. **Time** 5 min

Short Description Use the SAP Fiori Launchpad to create the outbound delivery note for the sales order.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

To create an outbound delivery, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Create Outbound Delivery*.

Start



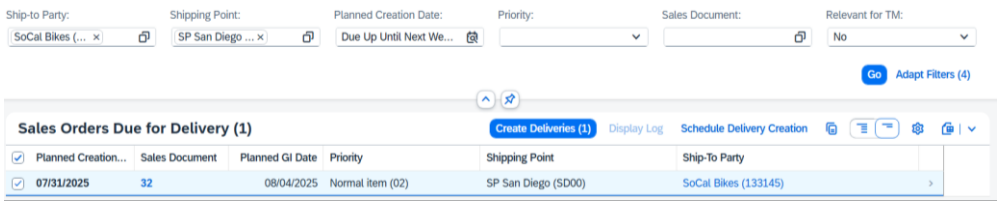
You will be taken to the *Create Outbound Delivery* screen. If you cannot see the header area, expand it by clicking on the button . In the header area that now opens, enter **133###** (SOCAL Bikes Irvine; replace ### with your number) as Ship-To Party, **SD00** as *Shipping point*, and **Due up until next week** as Planned Creation Date using the input help . Start the search by pressing Go.

133###

SD00

Due up until next week

In the *Create Outbound Deliveries* screen, select the Sales Document and click on the button Create Deliveries (1). You will receive a message that one log was created.



A log will then be generated. Click on the button Display Log to display the log. Make a note of the number that uniquely identifies your delivery.

Click on SAP to return to the SAP Fiori Launchpad.

Step 4: Change Outbound Delivery

Task Change the outbound delivery. **Time** 10 min

Short Description Use the SAP Fiori Launchpad to assign the storage location to the outbound delivery note for the sales order.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

To change the outbound delivery, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Change Outbound Delivery*.

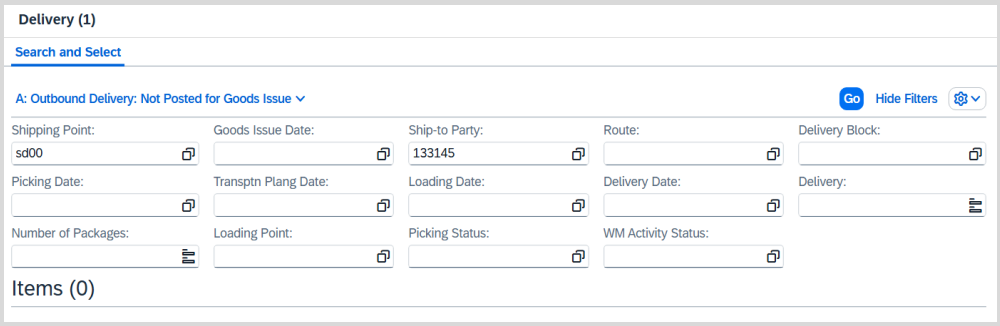
Start



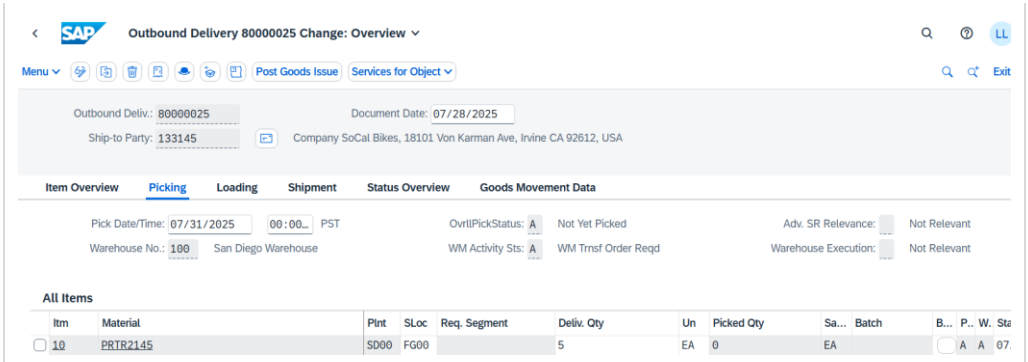
In the *Change Outbound Delivery* screen, enter your delivery number.

Note Alternatively, use the F4 help in the field *Outbound Delivery*. In the tab *Outbound Delivery: Not Posted for Goods Issue* enter **133###** (SOCAL Bikes Irvine; replace ### with your number) as Ship-To Party and **SD00** as Shipping point to find your Outbound Delivery number. Confirm your entries by pressing Go. Double-click on your outbound delivery to transfer it to the field Outbound Delivery.

133###
SD00



Confirm it with Continue to get to the *Outbound Delivery Change Screen*.



Enter your Storage Location **Finished Goods** and confirm with Enter. By filling the field Storage Location, the SAP System realizes that you selected a WM controlled warehouse. Click on Save to save your changes.

FG00



Click on SAP to return to the SAP Fiori Launchpad.

Step 5: Display Material Inventory

Task View the inventory of your material again. **Time** 5 min

Short Description Use the Fiori Launchpad to display the inventory of your material again.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

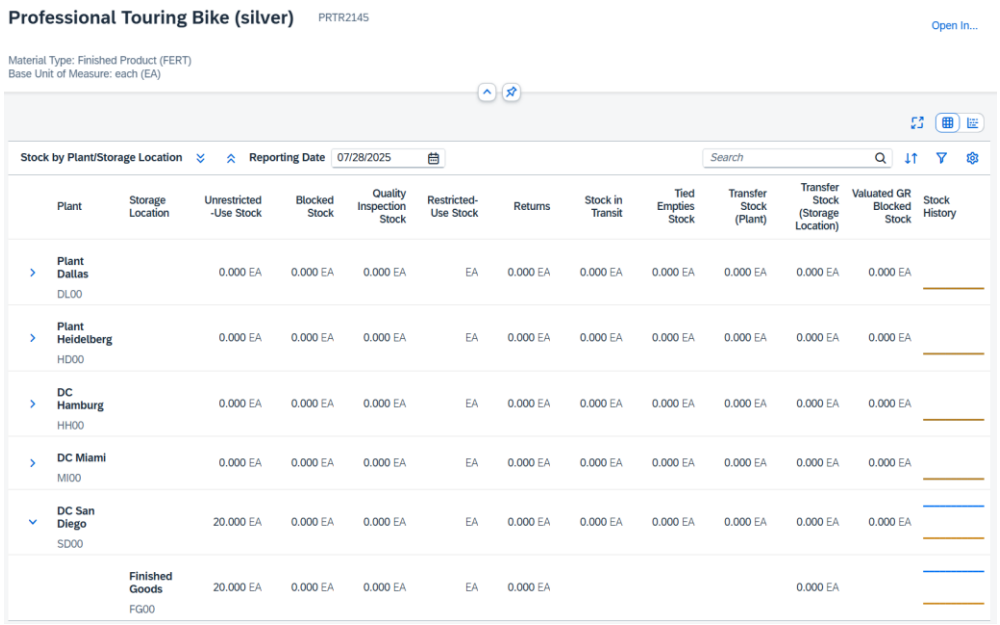
To display the material inventory, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Stock - Single Material.*

Start



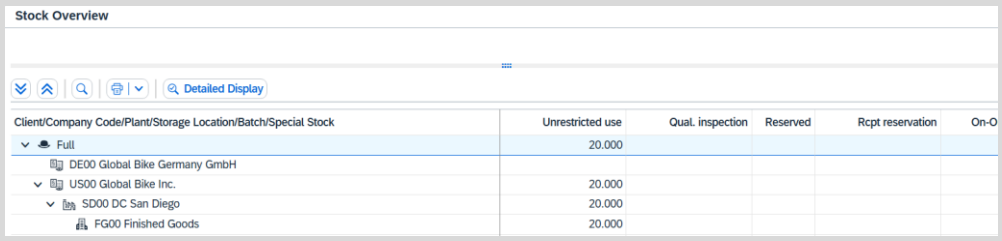
In the *Stock - Single Material* screen, enter **PRTR2###** as Material (remember to replace ### with your number). Klick Enter.

PRTR2###



Note that the amount of goods in San Diego has not changed yet.

Note In the App *Display Stock Overview* enter **PRTR2###** as Material. Select Execute to go to the stock overview. In the stock overview, you can scroll to the right to see the quantity (5) for the upcoming delivery.



Click on SAP to return to the SAP Fiori Launchpad.

Step 6: Pick Materials with Transfer Order

Task Create a transfer order to pick the materials. **Time** 10 min

Short Description Use the Fiori Launchpad to create a transfer order based on the delivery note created previously.

Name (Position) Sunil Gupta (Warehouse Employee)

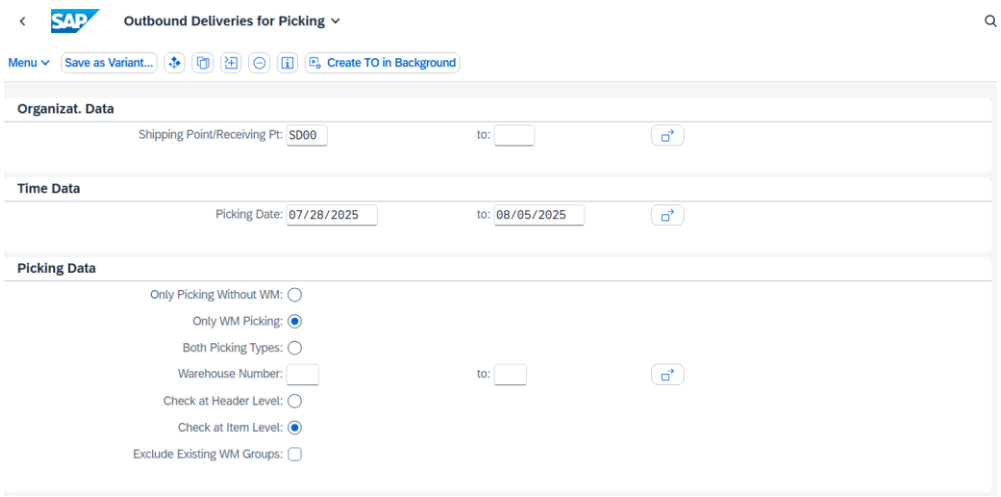
To create an outbound delivery, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Employee* role the app *Outbound Deliveries for Picking.*

Start



In the *Outbound Deliveries for Picking* screen, enter **SD00** as Shipping Point/Receiving Pt and select **Only WM Picking**.

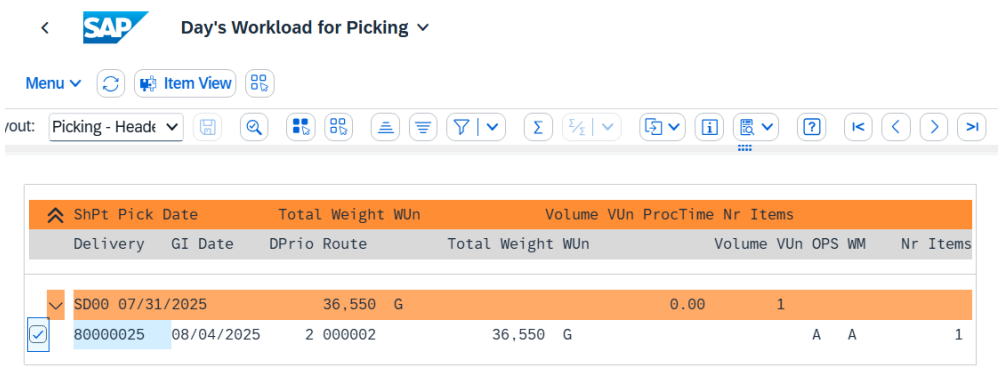
SD00
Only WM-Picking



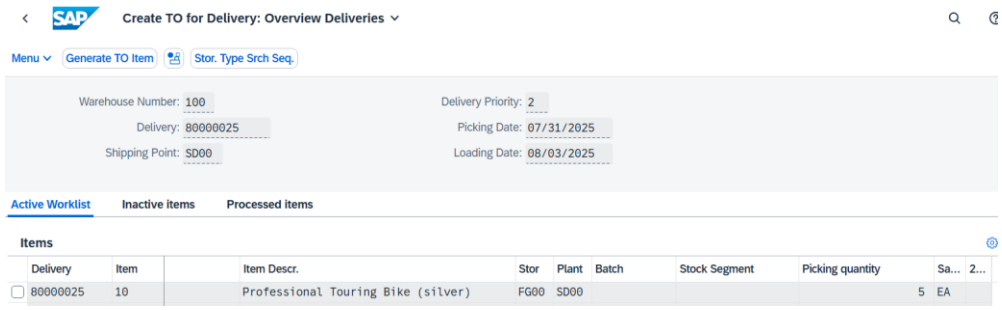
Then, click on Execute or press F8 to execute.

In the *Day's Workload for Picking* screen, you should see a line item with your **Outbound Delivery Number**. Select it and click on the TO in Foreground button.

Outbound Delivery Number

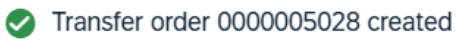


In the *Create Transfer Order for Delivery Note: Initial Screen*, press Enter to create your transfer order to be able to pick your documents. This will produce the following screen.



On the next screen, click on Posting to save your transfer order. The system will save the transfer order automatically and assign a unique transfer order number. Please note your number: _____

Transfer Order Number



Click on SAP to return to the SAP Fiori Launchpad.

Step 7: Run Bin Status Report

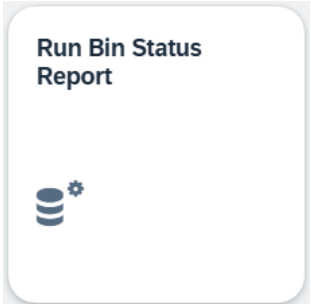
Task Check the status of your bins. **Time** 5 min

Short Description Use the Fiori Launchpad to run a bin status report, which will display a detailed report of each storage bin within the specified warehouse.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

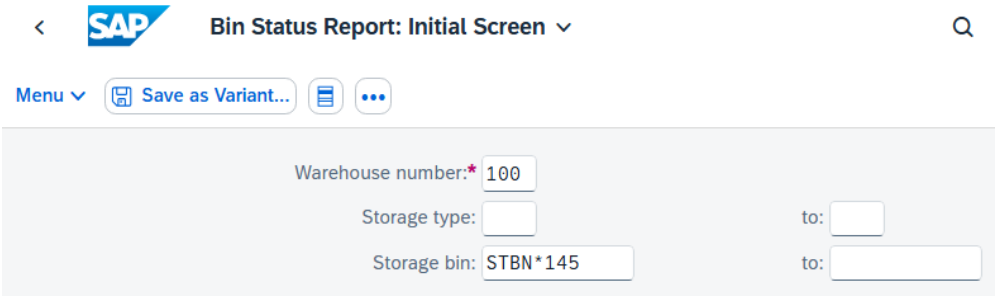
To run a bin status report, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Run Bin Status Report.*

Start

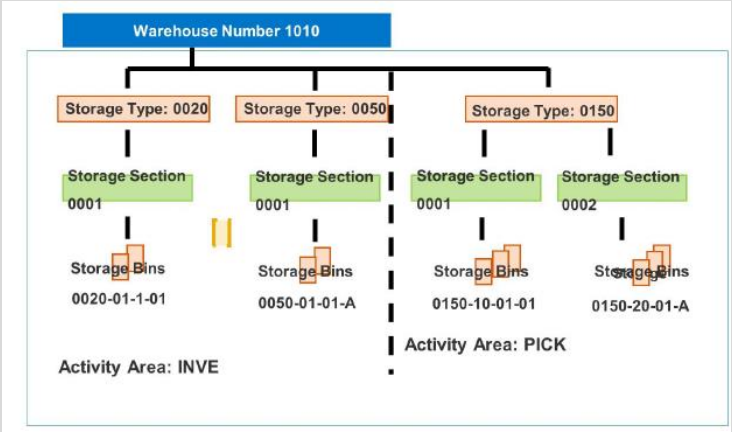


In the *Bin Status Report: Initial Screen*, enter **100** (for your San Diego Warehouse) as Warehouse Number and **STBN*###** as Storage bin (replace ### with your number). Then, click on Execute.

100
STBN*###



Note The warehouse number is the highest level of organizational unit in warehouse management. In practice, the warehouse number usually corresponds to a physical building or distribution center. Each warehouse number has a substructure that maps the spatial relationship in the warehouse complex in detail.



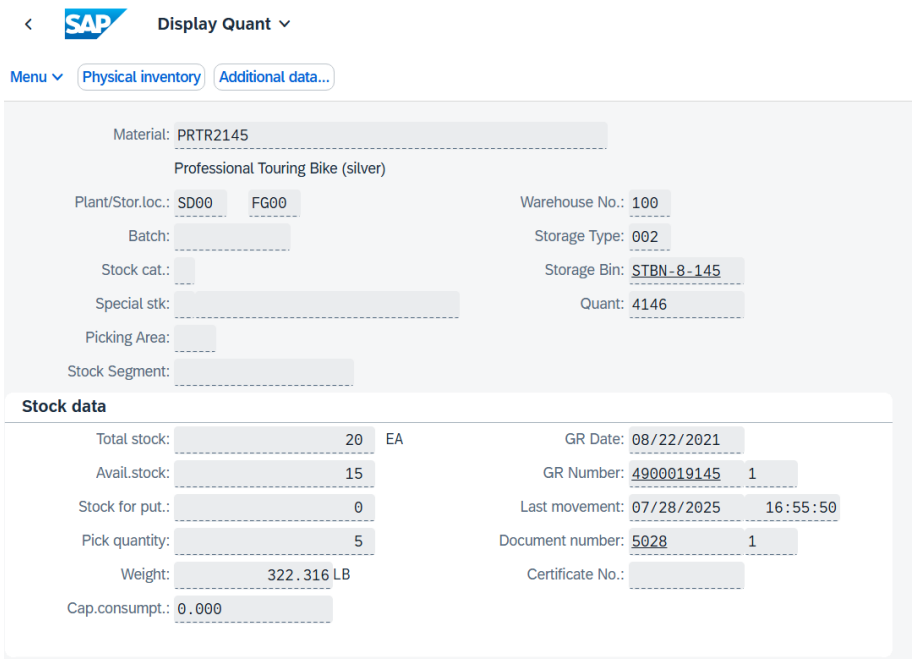
Storage bins are the lowest level of organizational structure. They are assigned to a storage type and a storage section (if one exists). Storage bins represent the physical location, where the goods are stored in the warehouse.

In the *Bin Status Report: Overview* screen you should see that the Storage Bin **STBN-8**-### is filled.

Typ	StorageBin	Material	Plnt	B
001	STBN-1-145	<<empty>>		
001	STBN-2-145	<<empty>>		
001	STBN-3-145	<<empty>>		
002	STBN-7-145	<<empty>>		
002	STBN-8-145	PRTR1145	SD00	
002	STBN-8-145	PRTR2145	SD00	
002	STBN-9-145	ORMN1145	SD00	
002	STBN-9-145	ORWN1145	SD00	

Click on your material to display detailed information of this quant and check whether 10 pieces of your good are stored in it.

Note As you can see 15 pieces of your material are marked as available stock and the other 5 as pick quantity.



Click on SAP to return to the SAP Fiori Launchpad.

Step 8: Confirm Transfer Order

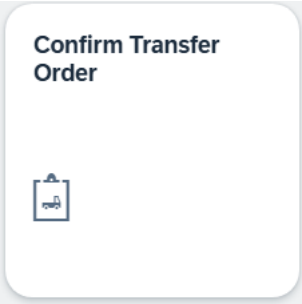
Task Confirm your transfer order. **Time** 5 min

Short Description Use the Fiori Launchpad to confirm the transfer order, you created in the previous step. This confirms that the goods have been physically picked from the storage bin, indicated in the transfer order.

Name (Position) Sunil Gupta (Warehouse Employee)

To confirm a transfer order, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Employee* role the app *Confirm Transfer Order*.

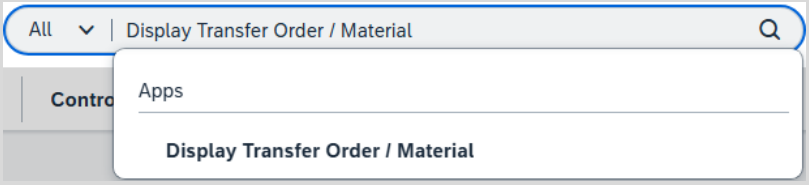
Start



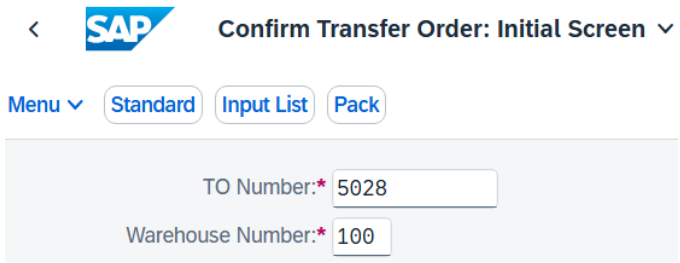
In the *Confirm Transfer Order: Initial Screen*, enter the **TO Number** from the previous task and **100** as Warehouse Number. Then, press Enter.

TO Number
100

Note If you have not written down the number, you can search for it using the app *Display Transfer Order*.

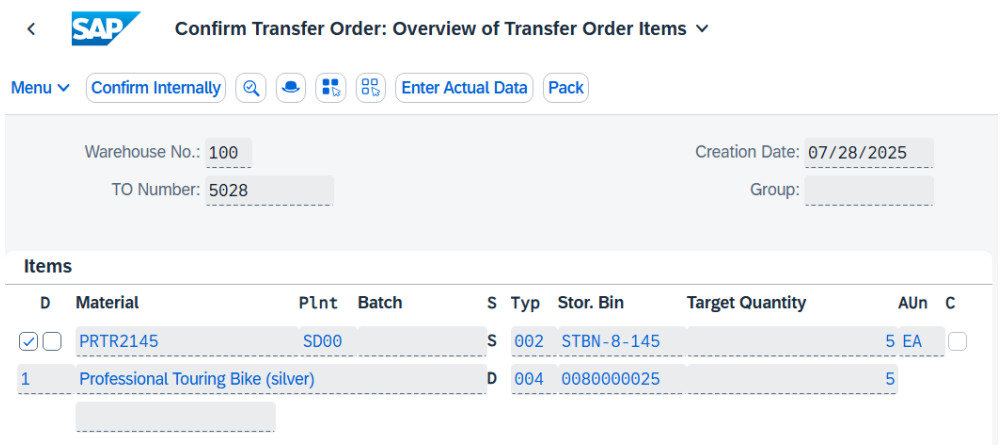


In the *Transfer Orders: List of Resident Documents* you have to fill in **100** as *Warehouse number*. Then, click on Execute.

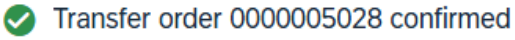


In the *Confirm Transfer Order: Overview of Transfer Order Items* screen, you should see an overview of your transfer order created in the previous step.

Review all of the details to make sure you have the correct quantity and storage bin.



Then, click on Posting to confirm your transfer order. The system will return a success message.



Click on SAP to return to the SAP Fiori Launchpad.

Step 9: Run Bin Status Report

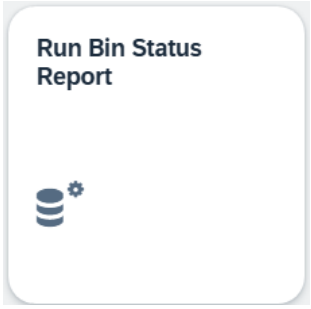
Task Check the status of your bins again. **Time** 5 min

Short Description Use the Fiori Launchpad to run a bin status report, which displays a detailed report of each storage bin within the specified warehouse.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

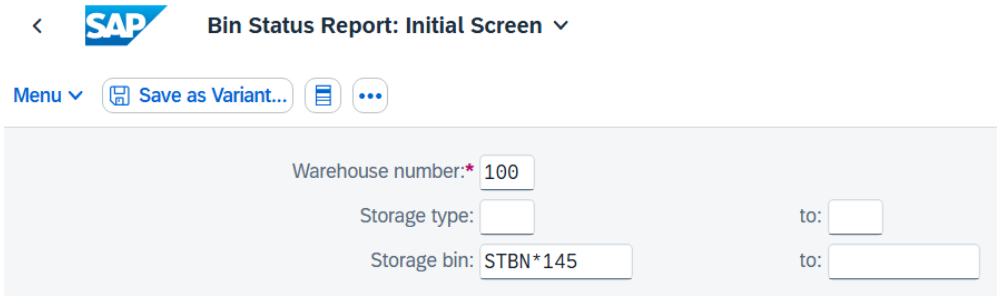
To run a bin status report, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Run Bin Status Report.*

Start

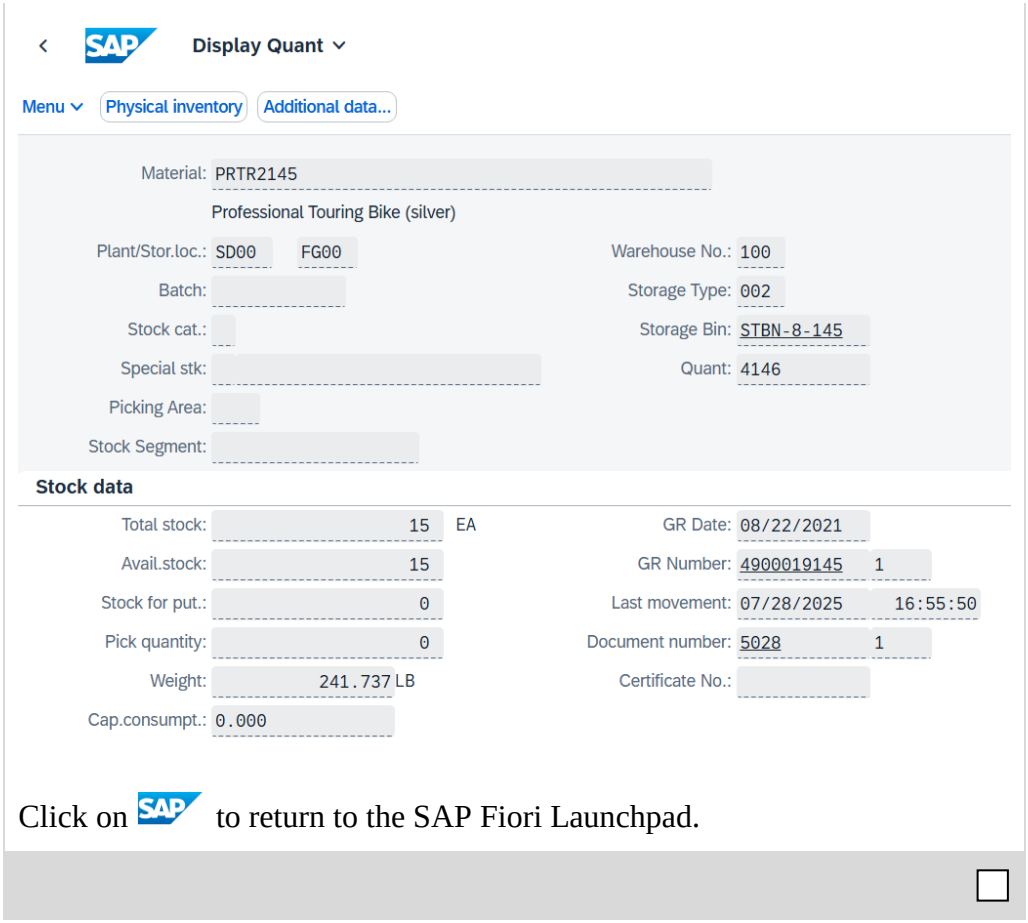


In the *Bin Status Report: Initial Screen*, enter **100** (for your San Diego Warehouse) as Warehouse Number and **STBN*###** as Storage bin (replace ### with your number). Then, click on Execute.

100
STBN*###



In the *Bin Status Report: Overview* screen click on your material in Storage Bin **STBN-8-###** to display detailed information and check if there are just 15 pieces of your good left.



Click on to return to the SAP Fiori Launchpad.

Step 10: Display Material Inventory Value

Task View the value of your material inventory. **Time** 5 min

Short Description Use the Fiori Launchpad to display the value of your material inventory.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

To display the material inventory value, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Display Warehouse Stock.*

Start



In the *Display Warehouse Stocks of Material* screen, enter **PRTR2###** as Material (replace ### with your number) and **SD00** as Plant. Ensure that all other search criteria fields are blank and click on Execute.

PRTR2###
SD00

Material		Material Description		Plnt Name 1		
Locat SL	Unrestricted Unit	Transit/Transf.	Quality Inspection	Restricted-Use	Blocked Stock	Returns
	Total Value Crcy	Total Value	Total Value	Total Value	Total Value	Total Value
PRTR2145		Professional Touring Bike (silver)		SD00 DC San Diego		
FG00	20 EA	0	0	0	0	0
	30,000.00 USD	0.00	0.00	0.00	0.00	0.00
* Total						
	30,000.00 USD	0.00	0.00	0.00	0.00	0.00

Note the total value associated with the DC in San Diego.

Click on SAP to return to the SAP Fiori Launchpad.

Step 11: Ship Materials

Task Ship the materials. **Time** 10 min

Short Description Use the Fiori Launchpad to ship the materials by posting a goods issue. This will reduce unrestricted stock to reflect that the inventory is shipped. It also indicates a change of ownership.

Name (Position) Zarah Morello (Good Issue Clerk)

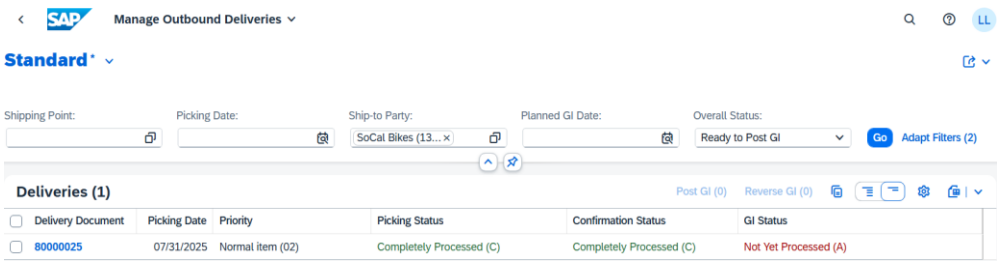
To ship materials, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Good Issue Clerk* role the app *Manage Outbound Deliveries.*

Start



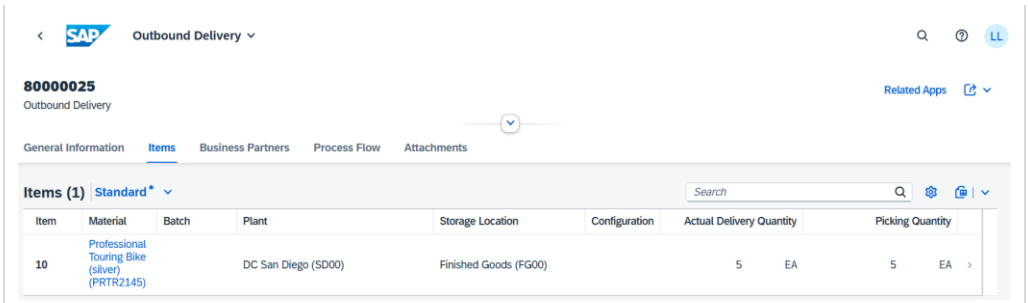
In the *Manage deliveries* view, you will see a list of all deliveries. To narrow down the list, expand the header area by clicking on the button . Enter your customer number **133###** in the *Ship-to Party* field. After you have initiated the search with Go, only your delivery will be displayed in the lower area.

133###

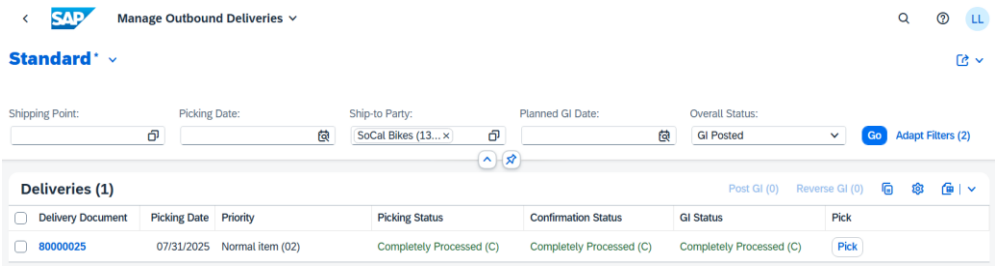


You can see that the outbound delivery has been fully picked and confirmed, but the goods issue is still pending.

In the *Outbound delivery* view, you will first see the general information. By clicking twice on the delivery document number and displaying the delivery, check the picked quantity 5 in the Items area.



After a successful check, click on ‹ to return to the *Manage outbound deliveries* view. In the *Deliveries* area, select your outbound delivery and then click on Post GI (1). In the popup *Post Goods Issue*, select today's date and click on Post GI (1) to confirm. The system will display a success message. If you want to check the status of the outbound delivery, you must adjust the *Overall status* field in the header area accordingly.



Click on SAP to return to the SAP Fiori Launchpad.

Step 12: Display Material Inventory

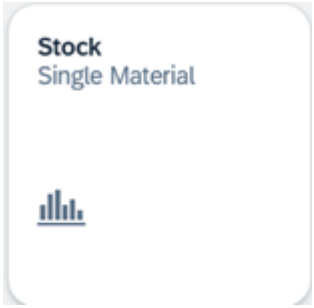
Task View the inventory of your material again. **Time** 5 min

Short Description Use Fiori Launchpad to display the inventory of your material again.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

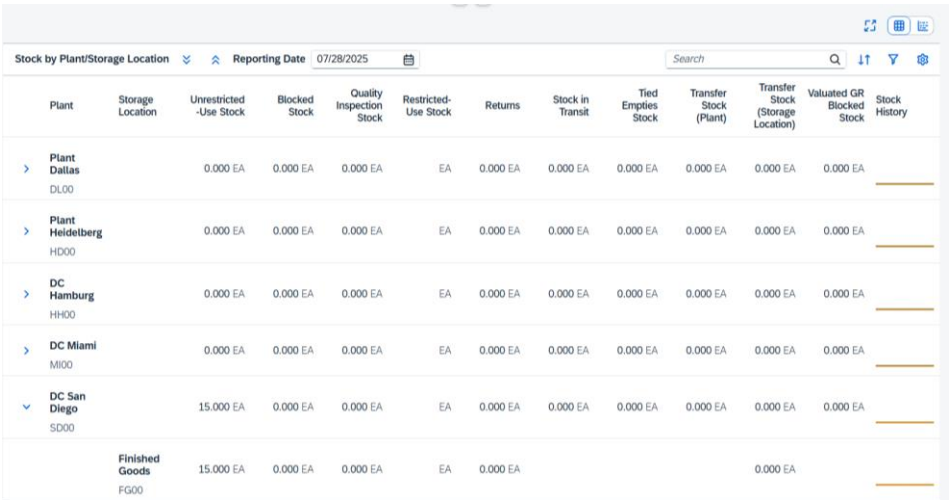
To display the material inventory, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Stock - Single Material.*

Start



In the *Stock - Single Material* screen, enter **PRTR2###** as Material (remember to replace ### with your number). Click Enter.

PRTR2###



Note that the amount of your goods in San Diego has decreased. This represents that 5 pieces of your material actually have been shipped.

Click on SAP to return to the SAP Fiori Launchpad.

Step 13: Display Material Inventory Value

Task View the value of your material inventory again. **Time** 5 min

Short Description Use the SAP Fiori Launchpad to display the value of your material inventory again.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

To display the material inventory value, use in the *Warehouse Management* area on the *Issue from Storage* page in the *Warehouse Supervisor* role the app *Display Warehouse Stock.*

Start



In the *Display Warehouse Stocks of Material* screen, enter **PRTR2###** as Material (remember to replace ### with your number) and **SD00** as Plant. Ensure that all other search criteria fields are blank and click on Execute.

PRTR2###
SD00

Material			Material Description		Plnt Name 1		
Locat SL	Unrestricted Unit		Transit/Transf.	Quality Inspection	Restricted-Use	Blocked Stock	Returns
	Total Value	Crcy	Total Value	Total Value	Total Value	Total Value	Total Value
PRTR2145			Professional Touring Bike (silver)		SD00 DC San Diego		
FG00	15	EA	0	0	0	0	0
	22,500.00	USD	0.00	0.00	0.00	0.00	0.00
* Total							
	22,500.00	USD	0.00	0.00	0.00	0.00	0.00

As you can see the values, associated with the material in San Diego, has been decreased. So there is no value associated to the *Transit/Transf.* for San Diego. This indicates the change of ownership after processing the shipping order.

Click on SAP to return to the SAP Fiori Launchpad.

WM III Challenge

Learning Objective Understand and perform a warehousing sales process cycle. **Time** 65 min

Motivation After having finished the *Warehouse Management III* case study successfully, you should now be able to solve the following challenge.

Scenario The warehouse management system has been tested without any problems, so the management decided to use the system productively. Due to a higher demand of bikes for the Tour de France, black Professional Touring Bikes are almost out of stock in Europe. They are still available in the Distribution Center in San Diego, so the delivery of the wholesale sales order can be managed by the new warehouse management system.

The customer VeloDOM from Magdeburg (Germany) ordered 20 black Professional Touring Bikes with a delivery time of 10 days.

Task Information You can use the case study *Warehouse Management III* as a guideline, but it is recommended to complete this challenge without further assistance to prove your WM skills.



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